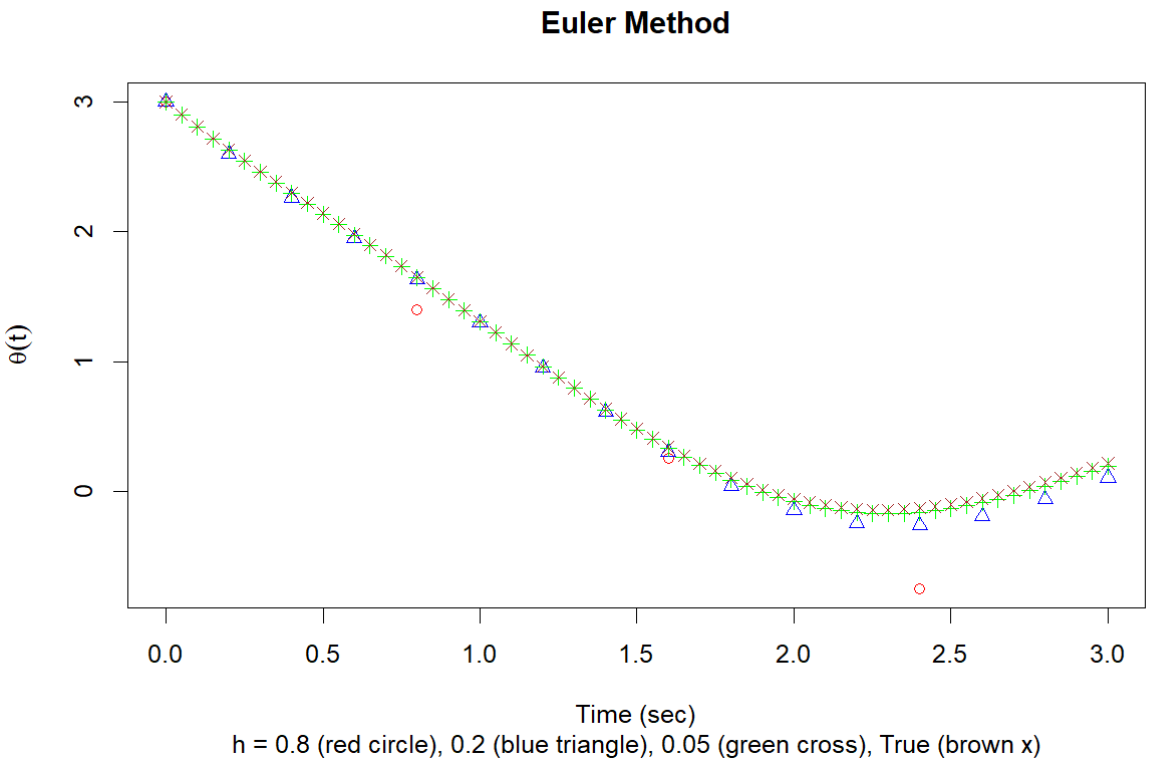
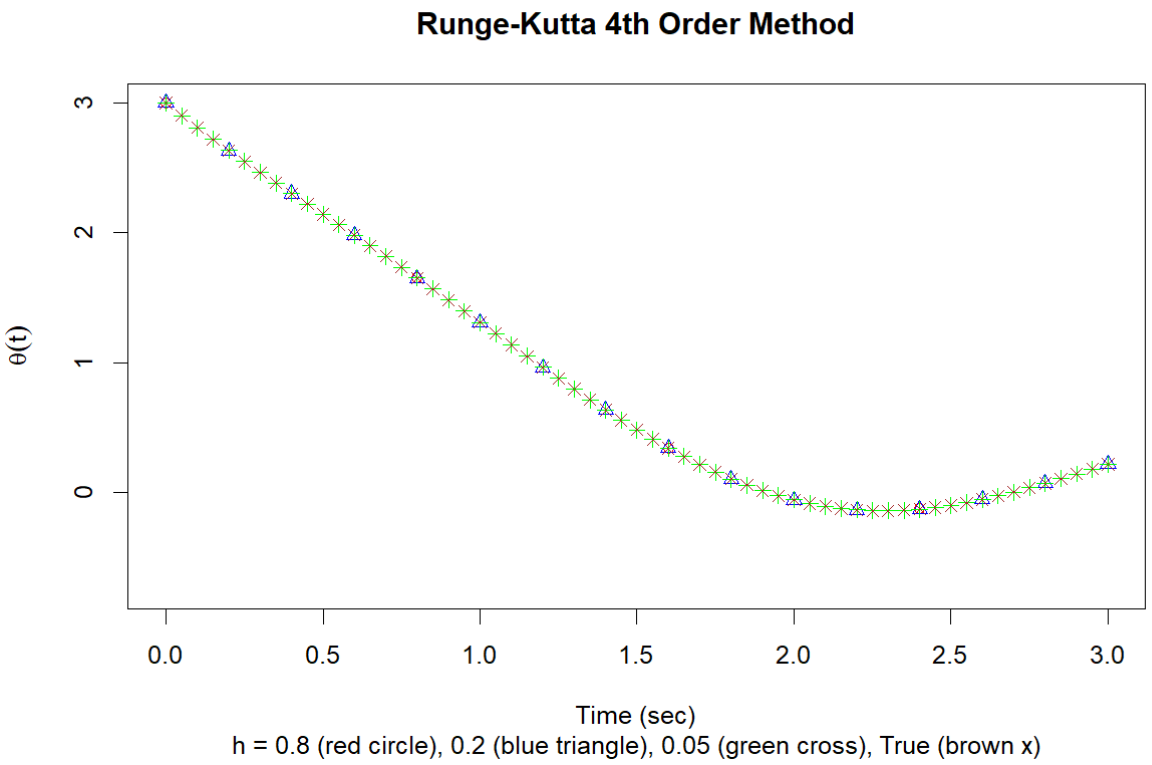
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PART-A

1. Which method has the highest accuracy?

   ➔ The Runge-Kutta 4th order method (RK4) has the highest accuracy.

2. Which values for h have the highest accuracy?
   ➔ The smallest step size, h = 0.05, has the highest accuracy.

3. What evidence exists to support your choices for the previous questions?
   ➔ The evidence comes from the percent error values and the graphs. The RK4 method has very small errors compared to Euler's method, which shows it is more accurate. Also, when the step size gets smaller, the errors become much smaller. In the graphs, RK4 points are very close to the true solution, while Euler points are farther away, especially for larger h. This proves that RK4 and smaller h give better accuracy.

PART-B

1. What is a differential equation?
   ➔ A differential equation is an equation that includes a function and its derivative. It shows how a quantity changes with respect to another variable, usually time.

2. What makes an ordinary differential equation, ordinary?
   ➔ An ordinary differential equation (ODE) is called ordinary because it involves derivatives with respect to only one independent variable.

3. What is the purpose of the Runge-Kutta methods of numerical analysis?
   ➔ The purpose of the Runge-Kutta methods is to find accurate approximate solutions to differential equations when an exact solution is difficult or impossible to find. They improve accuracy by using multiple calculations within each step instead of just one, like in Euler's method.

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```
> print(euler_h1)
  time     approx       true percent_error
1  0.0  3.0000000  3.0000000            NA
2  0.8  1.4000000  1.6521106      15.25991
3  1.6  0.2566404  0.3423016      25.02508
4  2.4 -0.7473077 -0.1269558     488.63626
> print(euler_h2)
   time      approx        true percent_error
1   0.0  3.00000000  3.00000000            NA
2   0.2  2.60000000  2.63242564     1.2317782
3   0.4  2.26421220  2.30317991     1.6919090
4   0.6  1.95071110  1.98195977     1.5766549
5   0.8  1.63304043  1.65211064     1.1542934
6   1.0  1.30059244  1.31055204     0.7599544
7   1.2  0.95724459  0.96605032     0.9115195
8   1.4  0.61831693  0.63602229     2.7837650
9   1.6  0.30620907  0.34230164    10.5440821
10  1.8  0.04530830  0.10647703    57.4478139
11  2.0 -0.14310504 -0.05451093   162.5254115
12  2.2 -0.24521276 -0.13185856    85.9665061
13  2.4 -0.25763678 -0.12695578   102.9342694
14  2.6 -0.18860963 -0.05171251   264.7272791
15  2.8 -0.05718437  0.07287470   178.4694423
16  3.0  0.10936568  0.21967165    50.2140208
> print(euler_h3)
   time       approx         true percent_error
1  0.00  3.000000000  3.000000000            NA
2  0.05  2.900000000  2.902376588    0.08188422
3  0.10  2.804750208  2.809025818    0.15220971
4  0.15  2.713516027  2.719257714    0.21114908
5  0.20  2.625607050  2.632425644    0.25902325
6  0.25  2.540379747  2.547928920    0.29628665
7  0.30  2.457239888  2.465215130    0.32351101
8  0.35  2.375644674  2.383782164    0.34136883
9  0.40  2.295104550  2.303179907    0.35061774
10 0.45  2.215184658  2.223011582    0.35208653
11 0.50  2.135505923  2.142934702    0.34666380
12 0.55  2.055745742  2.062661637    0.33528984
13 0.60  1.975638261  1.981959766    0.31895223
14 0.65  1.894974236  1.900651215    0.29868598
```

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15	0.70	1.813600466	1.818612171	0.27557858
16	0.75	1.731418800	1.735771783	0.25078084
17	0.80	1.648384720	1.652110636	0.22552464
18	0.85	1.564505508	1.567658835	0.20114882
19	0.90	1.479838007	1.482493681	0.17913555
20	0.95	1.394486002	1.396736987	0.16116026
21	1.00	1.308597224	1.310552039	0.14915965
22	1.05	1.222360021	1.224140223	0.14542471
23	1.10	1.135999715	1.137737372	0.15272926
24	1.15	1.049774673	1.051609835	0.17450977
25	1.20	0.963972138	0.966050322	0.21512173
26	1.25	0.878903846	0.881373566	0.28021264
27	1.30	0.794901473	0.797911819	0.37727803
28	1.35	0.712311961	0.716010253	0.51651387
29	1.40	0.631492756	0.636022291	0.71216606
30	1.45	0.552807001	0.558304905	0.98474938
31	1.50	0.476618743	0.483213952	1.36486320
32	1.55	0.403288181	0.411099562	1.90011895
33	1.60	0.333167014	0.342301638	2.66858889
34	1.65	0.266593925	0.277145512	3.80723741
35	1.70	0.203890240	0.215937788	5.57917537
36	1.75	0.145355819	0.158962413	8.55963014
37	1.80	0.091265193	0.106477026	14.28649328
38	1.85	0.041864013	0.058709603	28.69307470
39	1.90	-0.002634189	0.015855435	116.61379506
40	1.95	-0.042050866	-0.021925524	91.78955756
41	2.00	-0.076244938	-0.054510929	39.87091875
42	2.05	-0.105114872	-0.081817904	28.47416882
43	2.10	-0.128600325	-0.103804474	23.88707378
44	2.15	-0.146683350	-0.120470567	21.75866145
45	2.20	-0.159389141	-0.131858560	20.87887291
46	2.25	-0.166786328	-0.138053378	20.81292755
47	2.30	-0.168986801	-0.139182144	21.41413833
48	2.35	-0.166145088	-0.135413382	22.69473354
49	2.40	-0.158457266	-0.126955778	24.81296194
50	2.45	-0.146159454	-0.114056529	28.14650354
51	2.50	-0.129525863	-0.096999277	33.53281309
52	2.55	-0.108866460	-0.076101660	43.05398981
53	2.60	-0.084524250	-0.051712509	63.45029817
54	2.65	-0.056872204	-0.024208713	134.92452458

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```
54 2.65 -0.056872204 -0.024208713  134.92452458
55 2.70 -0.026309877  0.006008216  537.89833483
56 2.75  0.006740261  0.038515836   82.50002792
57 2.80  0.041836736  0.072874696   42.59085995
58 2.85  0.078523193  0.108632438   27.71662450
59 2.90  0.116332673  0.145328048   19.95167162
60 2.95  0.154792015  0.182496197   15.18068991
61 3.00  0.193426336  0.219671649   11.94751969
>
> print(rk4_h1)
  time     approx       true percent_error
1  0.0  3.0000000  3.0000000            NA
2  0.8  1.6562457  1.6521106     0.2502896
3  1.6  0.3461211  0.3423016     1.1158284
4  2.4 -0.1212113 -0.1269558     4.5247804
> print(rk4_h2)
   time      approx        true percent_error
1   0.0  3.00000000  3.00000000            NA
2   0.2  2.63243007  2.63242564  0.0001680016
3   0.4  2.30318656  2.30317991  0.0002889002
4   0.6  1.98196755  1.98195977  0.0003927281
5   0.8  1.65211924  1.65211064  0.0005206555
6   1.0  1.31056164  1.31055204  0.0007323539
7   1.2  0.96606130  0.96605032  0.0011364880
8   1.4  0.63603500  0.63602229  0.0019982088
9   1.6  0.34231621  0.34230164  0.0042559603
10  1.8  0.10649326  0.10647703  0.0152414887
11  2.0 -0.05449359 -0.05451093  0.0318032061
12  2.2 -0.13184097 -0.13185856  0.0133387243
13  2.4 -0.12693898 -0.12695578  0.0132310082
14  2.6 -0.05169758 -0.05171251  0.0288771256
15  2.8  0.07288683  0.07287470  0.0166490874
16  3.0  0.21968034  0.21967165  0.0039548879
```

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```
> print(rk4_h3)
   time       approx         true percent_error
1  0.00  3.000000000  3.000000000           NA
2  0.05  2.902376593  2.902376588  1.494992e-07
3  0.10  2.809025826  2.809025818  2.868555e-07
4  0.15  2.719257726  2.719257714  4.131037e-07
5  0.20  2.632425658  2.632425644  5.294787e-07
6  0.25  2.547928937  2.547928920  6.373960e-07
7  0.30  2.465215148  2.465215130  7.384321e-07
8  0.35  2.383782183  2.383782164  8.343087e-07
9  0.40  2.303179928  2.303179907  9.268808e-07
10 0.45  2.223011605  2.223011582  1.018131e-06
11 0.50  2.142934726  2.142934702  1.110169e-06
12 0.55  2.062661662  2.062661637  1.205244e-06
13 0.60  1.981959792  1.981959766  1.305761e-06
14 0.65  1.900651242  1.900651215  1.414310e-06
15 0.70  1.818612199  1.818612171  1.533714e-06
16 0.75  1.735771812  1.735771783  1.667082e-06
17 0.80  1.652110666  1.652110636  1.817893e-06
18 0.85  1.567658866  1.567658835  1.990100e-06
19 0.90  1.482493713  1.482493681  2.188264e-06
20 0.95  1.396737021  1.396736987  2.417738e-06
21 1.00  1.310552074  1.310552039  2.684912e-06
22 1.05  1.224140260  1.224140223  2.997533e-06
23 1.10  1.137737411  1.137737372  3.365157e-06
24 1.15  1.051609875  1.051609835  3.799767e-06
25 1.20  0.966050364  0.966050322  4.316655e-06
26 1.25  0.881373609  0.881373566  4.935700e-06
27 1.30  0.797911864  0.797911819  5.683266e-06
28 1.35  0.716010301  0.716010253  6.595103e-06
29 1.40  0.636022340  0.636022291  7.720931e-06
30 1.45  0.558304956  0.558304905  9.131958e-06
31 1.50  0.483214005  0.483213952  1.093378e-05
32 1.55  0.411099616  0.411099562  1.328979e-05
33 1.60  0.342301694  0.342301638  1.646648e-05
34 1.65  0.277145571  0.277145512  2.092929e-05
35 1.70  0.215937848  0.215937788  2.756893e-05
36 1.75  0.158962474  0.158962413  3.832716e-05
37 1.80  0.106477089  0.106477026  5.838516e-05
38 1.85  0.058709666  0.058709603  1.077095e-04
```

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```
39 1.90  0.015855499  0.015855435  4.043740e-04
40 1.95 -0.021925459 -0.021925524  2.954953e-04
41 2.00 -0.054510864 -0.054510929  1.196867e-04
42 2.05 -0.081817838 -0.081817904  8.001071e-05
43 2.10 -0.103804409 -0.103804474  6.304226e-05
44 2.15 -0.120470502 -0.120470567  5.409369e-05
45 2.20 -0.131858496 -0.131858560  4.901926e-05
46 2.25 -0.138053314 -0.138053378  4.624637e-05
47 2.30 -0.139182082 -0.139182144  4.511486e-05
48 2.35 -0.135413320 -0.135413382  4.540117e-05
49 2.40 -0.126955718 -0.126955778  4.719059e-05
50 2.45 -0.114056471 -0.114056529  5.093474e-05
51 2.50 -0.096999221 -0.096999277  5.777174e-05
52 2.55 -0.076101606 -0.076101660  7.063477e-05
53 2.60 -0.051712458 -0.051712509  9.911838e-05
54 2.65 -0.024208665 -0.024208713  2.005952e-04
55 2.70  0.006008262  0.006008216  7.604134e-04
56 2.75  0.038515879  0.038515836  1.107442e-04
57 2.80  0.072874735  0.072874696  5.418026e-05
58 2.85  0.108632475  0.108632438  3.332300e-05
59 2.90  0.145328081  0.145328048  2.258775e-05
60 2.95  0.182496226  0.182496197  1.610420e-05
61 3.00  0.219671675  0.219671649  1.179747e-05
>
```